Modelling and control of multibody mechanical systems

S. Evangelou

COURSE-WORK

Modelling and Control of Wheel Shimmy

1 Software details

To complete the coursework you will need to make use of the software tool, SimMechanics, developed by MathWorks.

> "SimMechanics™ provides a multibody simulation environment for 3D mechanical systems, such as robots, vehicle suspensions, construction equipment, and aircraft landing gear. You model the multibody system using blocks representing bodies, joints, constraints, and force elements, and then SimMechanics formulates and solves the equations of motion for the complete mechanical system. Models from CAD systems, including mass, inertia, joint, constraint, and 3D geometry, can be imported into SimMechanics. An automatically generated 3D animation lets you visualize the system dynamics.
>
> You can parameterize your models using MATLAB® variables and expressions, and design control systems for your multibody system in Simulink®. You can add electrical, hydraulic, pneumatic, and other components to your mechanical model using Simscape™ and test them all in a single simulation environment."

SimMechanics is installed on a Windows Server and can be accessed via a remote desktop connection from any windows, mac or unix machine, either inside the College, at home, or even from abroad. To log on to the server and access SimMechanics execute the following steps:

1. Write an email to `s.evangelou@imperial.ac.uk` with subject "capwints-msc <login name>" in which you write your own IC login name, to get permission to use the server. It may take 24 hours for your request to be processed so be patient.

2. Find a windows, mac or unix desktop machine inside your department or use your own PC or laptop at home by getting it connected to the Imperial College VPN (Virtual Private Network).

3. If you use a windows machine go through Start→Programs→Accessories→Communications and execute "Remote Desktop Connection". These steps may vary depending on the version of Windows you use. For unix machines use a remote desktop client such as Remmina or equivalent. For mac use Microsoft Remote Desktop.

4. Connect to the computer with hostname `capwints-msc.ee.ic.ac.uk`.

5. SimMechnics can be accessed by starting Matlab and typing 'simmechanics' on the command line. Alternatively, you may double-click on a simmechanics model file with extension 'slx'.

An introduction to SimMechanics is provided in the 'Getting Started Guide' found at `http://cn.mathworks.com/help/pdf_doc/physmod/sm/sm_gs.pdf`. Beyond this, there is plenty of documentation available on-line for every aspect of the software and you should not have any difficulty accessing it. The recommended approach is to open the model template that is provided to you (see below), double-click on each block and read its description. You may then follow up anything which is unclear by searching on-line. Note that we will be using the second generation of SimMechanics which is different from the older first generation.

2 Introduction

Caster wheel shimmy can occur in everyday equipment such as grocery trolleys, gurneys, and wheelchairs. These self-excited oscillations are an important consideration in the design of aircraft landing gear and road vehicle suspension and steering systems.

Ensuring the stability of wheel shimmy modes in aircraft landing gear, automotive steering systems, and single-track vehicles is vital due to the potential violence of the oscillations in these contexts. An idea of how instability arises can be obtained by examining simple cases–we will attempt to investigate one such simple case in this coursework–, but systems of practical importance are sufficiently complex to demand analysis by automated multibody modelling tools and numerical methods.

A simple system quite commonly employed to demonstrate wheel shimmy, both experimentally and theoretically is shown in Figure 1.

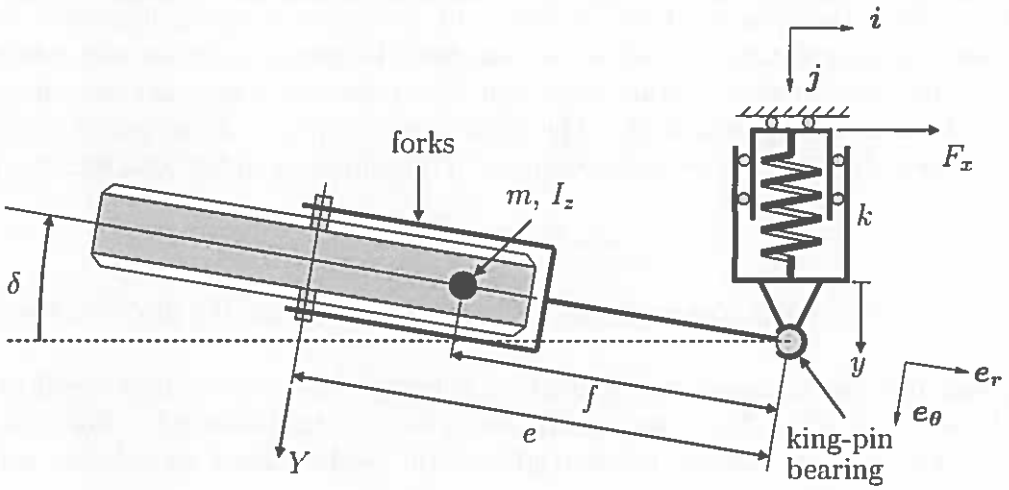


Figure 1: Plan (top) view of a simple system capable of shimmy.

The wheel is axisymmetric and free to spin relative to the forks that support it; the wheel is deemed to have no spin inertia. The wheel has mechanical trail e and mass offset f with respect to the vertical king-pin bearing. The king-pin is free to translate laterally with displacement y from static equilibrium, while the whole assembly moves forward with displacement x under the influence of a longitudinal force F_x. The king-pin mounting has stiffness k, while the moving assembly has mass m and z-moment of inertia I_z . The steer angle is δ. The king-pin is assumed massless so that analysis deals with only one body. The tyre-ground contact can be treated on one of three different levels. First, pure (nonholonomic) rolling, implying no sideslip, can be assumed. Second, the tyre may be allowed to sideslip thereby producing a proportionate and instantaneous side force. Third, the side force may be lagged relative to the sideslip by a first-order lag determined by the tyre relaxation length.

3 Tyre modelling

A tyre operates by generating forces at the interface between the elastic tyre tread base and the ground. The mechanisms by which these forces are generated are complicated but they have been studied extensively in the literature and a lot is already known. Under certain

circumstances we will assume that these apply in our current problem the lateral force developed by the tyre is proportional to the sideslip and acts to oppose the slip. The constant of proportionality is called the cornering stiffness.

Sideslip is defined as the lateral velocity of the tyre at the point of contact with the ground, divided by the absolute value of the longitudinal velocity of the wheel:

$$\alpha = \frac{v_{lat}}{|v_{long}|}.$$

In SimMechanics these two velocity components can be found by using a Transform Sensor.

When the tyre operates under transient conditions, following for example a step change in steering angle, the distortion of the tyre tread material does not develop instantly. Instead, the distortion builds up in a manner that is linked to the distance covered from the time of application of the transient. For vehicle modeling purposes, a simple approximation of this behavior is to treat the dynamic force development process as a speed-dependent first-order lag. The characterizing parameter, called the relaxation length σ, is similar to a time constant except that it has units of length rather than time. The relaxation length is a tyre characteristic that can be determined experimentally. The lateral force response of the tyre due to steering, and therefore side slipping, is a dynamic response to the slip angle of the tyre, which is modeled as

$$\frac{\sigma}{|v_{long}|}\dot{Y} + Y = -C\alpha.$$

The force Y acts in the lateral ($\boldsymbol{e_\theta}$ in Figure 1) direction and opposes the slip. C is the cornering stiffness.

In the case of a nonholonomic tyre the lateral velocity of the tyre is constrained to be zero. This condition is true when the cornering stiffness of the tyre tends towards infinity. In practice tyres have finite cornering stiffness but nevertheless the nonholonomic assumption is useful for analysis purposes.

4 Modelling, analysis and control tasks

You are asked to carry out certain modelling, analysis and control tasks of the system in Figure 1 and prepare a concise report with your results.

1. Assume that the tyre can sideslip and that the tyre force is lagged. Derive the equations of motion by hand and include the derivation and the equations in your report. Make use of a Cartesian and a polar coordinate system. The associated Cartesian unit vectors are $\boldsymbol{i}$, $\boldsymbol{j}$ and $\boldsymbol{k}$ as shown in Figure 1 where $\boldsymbol{k}$ is normal to the page with a direction along z inside the page, and the associated polar unit vectors are $\boldsymbol{e_r}$ and $\boldsymbol{e_\theta}$ and are attached onto the wheel assembly with their origin coinciding with the king-pin bearing.
[18 marks]

2. The forward speed of the king-pin, $\dot{x}$, is now constrained to have a constant value v by applying an appropriate force F_x. What should F_x be? Include the answer in your report.
[3 marks]

3. Assume that, in addition to the king-pin forward speed being constrained, the tyre (of the system in Figure 1) is subject to a nonholonomic constraint. Derive the equations of motion manually and include your hand derivation in your report. **[10 marks]**

The modelling and analysis tasks 4–9 and control task 10 involve the use of SimMechanics and Matlab. You will need to use the SimMechanics `coursework_model_template.slx` model template file, provided in the course Blackboard webpage, which includes default values for some of the system parameters. For the analysis tasks (6–9) and control task (10) you are encouraged to write all the Matlab commands in sequence in a single Matlab script, so that at the end you will be able to generate all the results by executing the script.

4. Build the model of the same system in 1, with the forward speed constraint in 2 also implemented, using SimMechanics. The file `coursework_model_template.slx` is essentially the model you need to build but it has been broken into four pieces. All you need to do is to connect up correctly the four pieces. Include a screen shot of the final SimMechanics model to show that you have connected up the model correctly. **[8 marks]**

5. By examining the model, write the default values for e, f, m, I_z and C. **[5 marks]**

6. Use the `linmod` command in Matlab to linearise your model in 4 as follows:

   ```
   [A,B,C,D] = linmod('coursework_model_template')
   ```

   Note that the linearisation should always be performed about $\delta = 0$. Calculate and plot on the complex plane the eigenvalues of the system (with imaginary parts equal or greater than zero only) with default parameter values (forward speed at 10 m/s and $\sigma = 0.2$) when the stiffness k of the king-pin mounting is varied from 5000 to 100000 N/m (use a `for` loop in Matlab with a step in k of 5000 say and call the `linmod` command in the loop). Include the plot in your report (all eigenvalues should be plot in one figure – imaginary part against real part). What can you say about the stability of the system as the spring stiffness varies? **[10 marks]**

7. Verify two operating cases by running two simulations as follows. Run a simulation for 4 seconds with default parameter values (forward speed at 10 m/s and $\sigma = 0.2$) except for k which should be set to 30000 N/m and the z rotational displacement of the forks initialised at 0.01 rad. Run a second simulation for 4 seconds with k set to 100000 N/m, the same initial condition on the forks' rotation and everything else left unchanged. Note that simulations can be run from the Matlab command line or in a script using the following command:

   ```
   sim('coursework_model_template')
   ```

   Plot the time response of the lateral displacement, y, of the king-pin and the z rotation of the forks for the two runs. Include the plots in your report. Does the result of the simulation agree with the result of the linear eigenvalue analysis in terms of stability interpretation, frequency of the response? Give your answer in the report. **[10 marks]**

8. Fix k to 100000 N/m. Sweep the forward speed from 1 to 50 m/s in steps of 1 m/s and calculate the eigenvalues of the linearised system for each speed. Plot the results on the complex plane close to the imaginary axis (real part > -20 say and imaginary parts equal or greater than zero) and include the plot in your report. What can you say about the dependence of shimmy oscillations with respect to forward velocity? **[8 marks]**

9. Fix k to 100000 N/m and the forward speed to 10 m/s. Vary the relaxation length, σ, from 0.02 to 1 m in steps of 0.02 m, calculate and plot the eigenvalues of the linearised system close to the imaginary axis (showing only eigenvalues with imaginary parts equal or greater than zero). Include this plot in the report. What can you say about the stability of the system with changes in σ? Verify the stability of the system by simulation when $\sigma = 1$ m (initial z rotation of the forks set to 0.01 rad). Include plots of the y king-pin displacement and z fork rotation for this case in your report. **[8 marks]**

The final task employs control to stabilise wheel shimmy and it involves the use of SimMechanics and Matlab. You will need to use the model you developed in 4 with default parameters (forward speed at 10 m/s and $\sigma = 0.2$) and with the stiffness of the king-pin mounting set to $k = 60000$ N/m.

10. Calculate and write the eigenvalues of the linearised system. Is the system unstable? Confirm your answer by initialising the z rotational displacement of the forks (δ) at 0.1 rad and running a simulation for 4 seconds. Plot the time response of the lateral displacement, y, of the king-pin and the z rotation of the forks, and include the plots in your report. How many unstable poles does the system have? (Note that, in the present case, eigenvalues at the complex origin correspond to uncontrollable modes that play no role in the dynamics of the system and they can be ignored).

    The system can be stabilised by the use of the following feedback control law:

$$T = -\lambda\dot{\delta}, \tag{1}$$

    where T is a torque applied on the wheel assembly about the steer axis and $\dot{\delta}$ is the steering velocity. Such a control law can be implemented in practice either by a passive (e.g. damper) or active (e.g. actuator) control system, but the value of λ needs to be determined first.

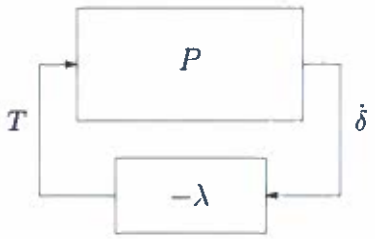


Figure 2: Feedback control law in (1) applied on the system P.

Modify your model by adding an input corresponding to T and an output corresponding to $\dot{\delta}$, using Simulink input and output ports, and linearise your (open-loop) model. Plot the Nyquist diagram from T to $\dot{\delta}$, using axes limits for which the Nyquist diagram fills up the biggest part of the plot. Include the plot in your report. Can you confirm the stability properties of the open loop system by inspecting the Nyquist plot? (You will need to recall the Nyquist stability criterion).

By using the Nyquist stability criterion determine the value of λ that stabilises the closed-loop system and write this value in your report. Implement the control law in your SimMechanics model, initialise the z rotational displacement of the forks at 0.1 rad and run a simulation for 4 seconds to confirm that the λ value you calculated produces marginal

stability. Plot the time response of the lateral displacement, y, of the king-pin and the z rotation of the forks, and include the plots in your report. You may want to run another simulation for a higher λ value (say, twice as much) to confirm that the closed-loop system will be stable. **[20 marks]**

Upload your report in pdf format on Blackboard by the deadline given on the course webpage.